Mobile application for the intelligent management of small shops using the GPT-4 model

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***Abstract*— Small shops and retail stores in Peru face significant challenges in efficient management due to limited access to technological resources, often relying on error-prone manual processes. This study presents the development of a mobile application powered by the GPT-4 model, designed to provide personalized recommendations and optimize key operations such as inventory management and sales forecasting. By leveraging GPT-4's advanced natural language processing capabilities, the application acts as an intelligent assistant, enabling shop owners to make informed decisions, automate routine tasks, and enhance business performance. Key features include product and sales management, dashboards, personalized recommendations, alerts, and other operational tools. A survey conducted with 15 shop owners revealed positive results, with over 66.7% rating the application between 4 and 5 on a Likert scale (1-5), indicating its effectiveness in improving small shop management with the support of GPT-4.**

Keywords*— *GPT-4, Management, Small Shops, Chatbot, Assistant

I. Introduction

In recent years, the management of small-scale retail establishments, such as small shops and stores, has posed significant challenges in the Peruvian context. These challenges stem from several factors, including limited access to technological resources, leading to a reliance on manual processes for inventory management, decision making, and performance evaluation [1]. The absence of efficient technological solutions exposes shop owners to errors in calculations, uninformed decision making, and suboptimal business performance [2]. This generates operational problems that negatively affect their performance, such as performing error-prone manual calculations for routine tasks, inventory control using rudimentary methods, and strategic decision making hampered by the lack of reliable data and analytical tools [3], [4], [5]. These limitations prevent small shop owners from realizing the full potential of their business and making informed decisions to improve their profitability and competitiveness in the marketplace.

This research aims to explore the potential of AI-driven approaches to improve the management of small shops in Peru. By leveraging the capabilities of GPT-4, the goal is to develop a mobile application that serves as an intelligent management tool for shop owners to manage their businesses, make informed decisions, and optimize their business operations. In that sense, with the integration of GPT-4, the application will provide personalized recommendations based on data from the small shop and an intelligent chatbot that provides instant access to information from the small shop, as well as serving as a management assistant.

II. Literature review

A. GPT and Large Language Model

The capabilities that AI technology contains are useful for various domains, in the case of the research focuses on the use of GPT's LLM model. For example, GPT demonstrates a robust ability to understand and generate natural language fluently. This is achieved thanks to its architecture of attention neural networks or "transformers" and the pre-training of 1 trillion parameters as in GPT-4 [6]. On the other hand, according to Keng-Boon's study, generative AI is being used in retail, so that it enables customers to have faster service and improved employee productivity [7]. In addition, Kurnia Muludi considers GPT as one of the best LLM models [8], because of the results she obtained in her research. Also, according to Amos Azaria, he highlights its ability to be used in scientific research, mathematics, programming, education and health care [9].

On the other hand, there are many LLM models, which can be open source or closed source. Each of these is trained with different data sizes, so the accuracy differs in each one. For this reason, Samuel Kernan's study compares different LLM models using different performance criteria [10]. These criteria are Factuality which refers to whether the generated answers are correct according to the context, Completeness which refers to whether the answers contain all the relevant information of the question and Hallucinations which refers to whether the answer is coherent according to the question. The study concludes that GPT-4 shows the best performance compared to other models. However, open-source models such as StableBeluga2 also show similar performances.

B. GPT applications for recommendation

GPT has various applications in the health sector. For instance, Ismael Dergaa's study utilized OpenAI's GPT-4 model to generate exercise routines tailored to individual goals. While the routines effectively supported fitness, they lacked deeper personalization [4]. Another notable use of GPT is in early disease detection. A study employing ChatGPT for the preliminary diagnosis of dengue fever reported highly satisfactory results [11]. Similarly, Li's research developed a chatbot leveraging GPT agents to evaluate responses to questions about traumatic brain injury rehabilitation [12]. Finally, Pash's study demonstrated how a GPT-powered chatbot, using patient-provided information, accurately identified the appropriate area of care [13].

The manufacturing sector has also benefited from the application of GPT. For example, in Kahiomba's study, a web application was developed to process data from production machines, enabling GPT to generate responses that help technicians identify issues quickly, thereby reducing

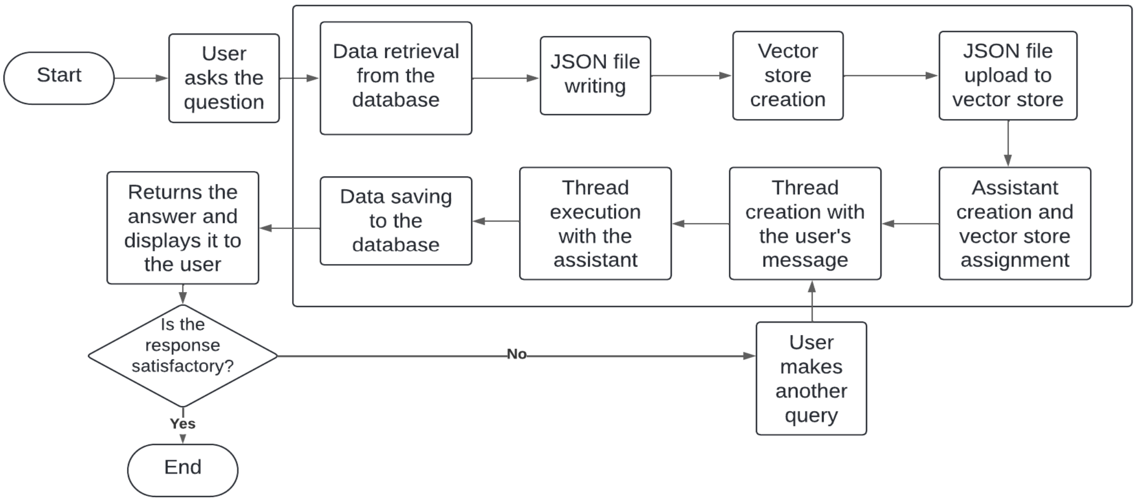

Fig. 1. Flowchart of user interaction and the GPT-4 model.

downtime [14]. Similarly, Wang's research pre-trained GPT on manufacturing datasets to create a system capable of addressing various tasks across different scenarios [15]. Additionally, GPT facilitates the generation of programming code for building graphs or models. Jackson's study demonstrates this by using GPT-3.5 to create simulation models of logistics systems in Python based on natural language prompts [3]. Likewise, Maddigan's research developed a web application powered by GPT-4 that not only generates Python code to construct graphs but also interprets and analyzes them through user queries [16].

III. Experimentation

A. Materials

The project aims to showcase the potential of a GPT-4-powered chatbot assistant for optimizing the management of small shops. To achieve this, an Android application will be developed, enabling shop owners to efficiently handle their businesses through intelligent recommendations and strategies generated by GPT-4. The app will facilitate key tasks such as product creation, supplier management, sales and purchase tracking, employee management, and dashboard visualization.

By leveraging data provided by shop owners and the GPT-4 model, the application will deliver personalized queries and responses, addressing topics such as shop performance and app usage. Additionally, GPT-4 will generate tailored business recommendations, including suggested products, offers, and discounts, empowering shop owners with actionable insights to gain a competitive edge in their markets.

The application will utilize MongoDB, a non-relational database, to store large volumes of data efficiently and with optimal latency. A REST API will be developed using NestJS, a Node.js framework that employs TypeScript and offers modular architecture, ensuring scalability and maintainability.

Furthermore, the application will integrate external services such as Cloudflare R2, OpenAI API, AWS SES, and Firebase Cloud Messaging for tasks like static file storage and push notifications. Cloud services will also be employed to host the API and database, specifically using Google Cloud's Compute Engine and MongoDB Atlas.

To evaluate the application's impact, a qualitative analysis will be conducted through a user survey involving 15 shop owners. This survey aims to assess the user experience and determine the application's effectiveness in improving shop management. Additionally, it will demonstrate the role of the GPT-4 model in enhancing business decision-making and operational efficiency.

B. Methodology

The project follows a clear methodology to implement the mobile application. The application, developed in Java 8, interacts with end users to manage various tasks such as creating products, managing suppliers, and tracking sales. The REST API, developed with NestJS, handles the business logic of the application and communicates with both the database and external services. MongoDB is used to store the data, and the OpenAI API interacts with the GPT-4 model to provide personalized recommendations and responses based on the shop's context.

The GPT-4 model contributes significantly to the research by enabling the creation of dynamic, context-aware recommendations for shop owners. When a user queries the system, the model processes the relevant data retrieved from the database, such as product sales and inventory details, and generates responses to help improve decision-making. The assistant's ability to interpret the data and provide tailored advice is critical for achieving the research objective of enhancing small shop management through artificial intelligence.

Additionally, Fig. 1 describes the process in which a user interacts with a GPT-based assistant. It begins when the user asks a question. The relevant data is retrieved from the database and saved, with a JSON file containing the obtained data. Then, a vector store is created, and the JSON file is uploaded to the vector store. A thread is created with the user's message, and the assistant executes the thread to process the query. If the response is satisfactory, the process ends; if not, the user can make another query, and the process repeats.

Fig. 2. Login.

Fig. 3. Home page.

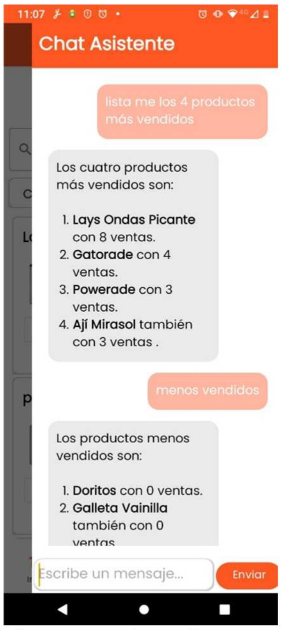

Fig. 4. Assistant chat.

Fig. 5. Add product.

A survey was conducted through Google forms to find out the perception of the shop owners about the proposed application. Fifteen shop owners participated in the survey and were provided with a link to download the application. The survey evaluated users' comfort and satisfaction with the application and GPT-4 chatbot. The results of the survey provided valuable information on the perceptions and experiences of the shop owners, which will enable the application to be improved and adapted to achieve greater acceptance and use in this market segment. The rating is based on a qualitative scale of 1 to 5, 1 "Strongly Disagree", 2 "Disagree", 3 "Neutral", 4 "Agree" and 5 "Strongly Agree".

C. *Results*

The initial page of the application is shown in Fig. 2, where the login process takes place. After the shop owner logs in, the "Home" page is displayed, showing the current day's sales data, including "Sales" and "Total" in soles. A small recommendation generated by the GPT-4 model, based on the shop's current data, is also provided to assist in decision-making. Additionally, options to manage employees, suppliers, offers and discounts, and purchases are available.

TABLE I. QUESTIONS AND RESPONSES OF THE SURVEY

No	Questions	Scale of responses				
		1	2	*3*	*4*	*5*
1	Is the application easy to navigate and find the desired functions?	0.0%	13.3%	33.3%	33.3%	20.0%
2	Does the application provide clear and concise information on inventory levels and stock status?	0.0%	13.3%	46.7%	33.3%	6.7%
3	Does the application generate accurate and timely reports on inventory movement and sales trends?	0.0%	6.7%	20.0%	40.0%	33.3%
4	Is the application's user interface intuitive and easy to use for small shop personnel?	0.0%	0.0%	13.3%	66.7%	20.0%
5	Does the application help to identify and prevent shortages and oversupply situations?	0.0%	33.3%	40.0%	26.7%	0.0%
6	Is product registration easy to perform?	0.0%	0.0%	6.7%	33.3%	60.0%
7	Do you consider the application useful to manage your small shop?	0.0%	0.0%	6.7%	60.0%	33.3%
8	Do you think that more data should be entered in the product registration?	0.0%	40%	46.7%	13.3%	0.0%
9	Is the application easy and user-friendly to install?	0.0%	0.0%	0.0%	20.0%	80.0%
10	Does the customer feel that their attention is faster because of the use of the application?	0.0%	13.3%	66.7%	20.0%	0.0%
11	Do you consider that making a sale is easy and friendly to do?	0.0%	0.0%	13.3%	53.3%	33.3%
12	Do you consider that you waste less time using the application when making a sale?	0.0%	0.0%	13.3%	46.7%	40.0%
13	Do you find the dashboard and graph sections where you can see how your small shop is performing useful?	0.0%	13.3%	60.0%	20.0%	6.7%
14	Do you consider the chatbot assistant useful?	0.0%	0.0%	0.0%	20.0%	80.0%
15	Do you feel that the chatbot's responses are accurate for your needs?	0.0%	0.0.%	0.0%	33.3%	66.7%

The chatbot can be accessed by swiping left from the right side of the screen, as shown in Fig. 3. Shop owners can use the chat to ask questions about their store or the application, as illustrated in Fig. 4.

Fig. 5 shows the new product registration view, which includes fields for name, purchase price, stock, price, unit, category, subcategory, supplier, image, and barcode. Barcodes are obtained from packaged products, while codes for unprocessed foods are generated and assigned.

Other modules in the application include dashboards, employee management, product management, supplier management, purchase management, offers and discounts, sales creation, profile management, password recovery, user registration, and recommendations.

All questions and responses are tabulated in Table 1. According to the survey, 8.89% of the results were in the 1-2 range, 24.44% were in the 3 range, and 66.67% were in the 4-5 range. These results indicate that the developed system is useful for shop owners. Question number 3 received 33.3%, which could be due to the complexity of an ERP (Enterprise Resource Planning). Additionally, GPT-4 has proven to be valuable in streamlining processes, improving decision-making, and providing actionable insights, which further enhances the system's utility for the shop owners.

IV. Conclusions

In this paper, the design and development of a mobile application for the intelligent management of small shops using the GPT-4 model has been presented. During the study, the system architecture was proposed and a detailed flow of interaction with the GPT-4 model was elaborated.

According to the survey of 15 shop owners and employees, the application proved to be useful, interactive, and easy to use. The GPT-4 model has enabled shop owners to query their shop data in an efficient and interactive way, significantly improving data accessibility and providing quick, accurate responses. The intuitive user interface has enhanced the user experience, reducing training time and boosting productivity. The chatbot's data accessibility and ease of use have been key factors in its success, promoting rapid adoption and efficient utilization.

In terms of limitations, our study was based on a limited number of small shops and operational scenarios. While the results were promising, the diversity of the shops evaluated does not fully represent all possible variations in the industry. Additionally, the application evaluation period was short, which may not have captured all the seasonal and operational variations a small shop may experience over time.

For future implications, incorporating IoT devices could improve product tracking and process automation in small shops, revolutionizing inventory management and logistics.

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